Flyweight

Season: 2 · Structural · Category: GoF · Level: Intermediate

Intent

- Share intrinsic state across many fine-grained objects.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Same glyph style reused for thousands of characters.

When to Use

- See Intent.
- The design force named in Intent shows up more than once.

When Not to Use

- When a simpler redesign removes the need.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.gof.structural.flyweight;

import java.util.Map;
import java.util.concurrent.ConcurrentHashMap;

record GlyphStyle(String font, int size, String color) {}

record Glyph(char character, GlyphStyle style, int x, int y) {}

/** Flyweight factory pools intrinsic glyph styles. */
public final class GlyphFactory {
    private final Map<GlyphStyle, GlyphStyle> pool = new ConcurrentHashMap<>();

    public GlyphStyle style(String font, int size, String color) {
        var key = new GlyphStyle(font, size, color);
        return pool.computeIfAbsent(key, k -> k);
    }

    public Glyph glyph(char c, GlyphStyle style, int x, int y) {
        return new Glyph(c, style, x, y);
    }

    public int pooled() { return pool.size(); }
}
```

Modern Java 25 / Functional

- Prefer sealed interfaces + records; virtual proxy uses synchronized lazy init.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: gof/structural/flyweight/
```

Pitfalls

- Keep adapters thin.
- Decorators should preserve the interface.
- Facades must not become god objects.

Interview Q&A

- Q: When is Flyweight the wrong choice? A: When you can change the adaptee/subject API or collapse the hierarchy.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Share intrinsic state across many fine-grained objects.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Same glyph style reused for thousands of characters. [0:40] Intent — State the design force in one s...